CS & IT ENGINEERING



Algorithms

Greedy Algorithms

Lecture No.- 05



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Optimal Merge Pattern.

Huffman Encoding

Topics to be Covered



- Topic
- Topic
- Topic

Huffman Encoding

Min Cost Spanning Trees (MCST)



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW



Topic : Greedy Algorithms

#Q. The characters 'A' to 'H' have the set of frequencies based on the first 8 Fibonacci number as follows:

A =1, B =1, C=2, D=3, E= 5, F= 8, G = 13, H=21

A Huffman code is used to represent the characters. What is the sequence of characters corresponding to the following code: 1101111001101 0

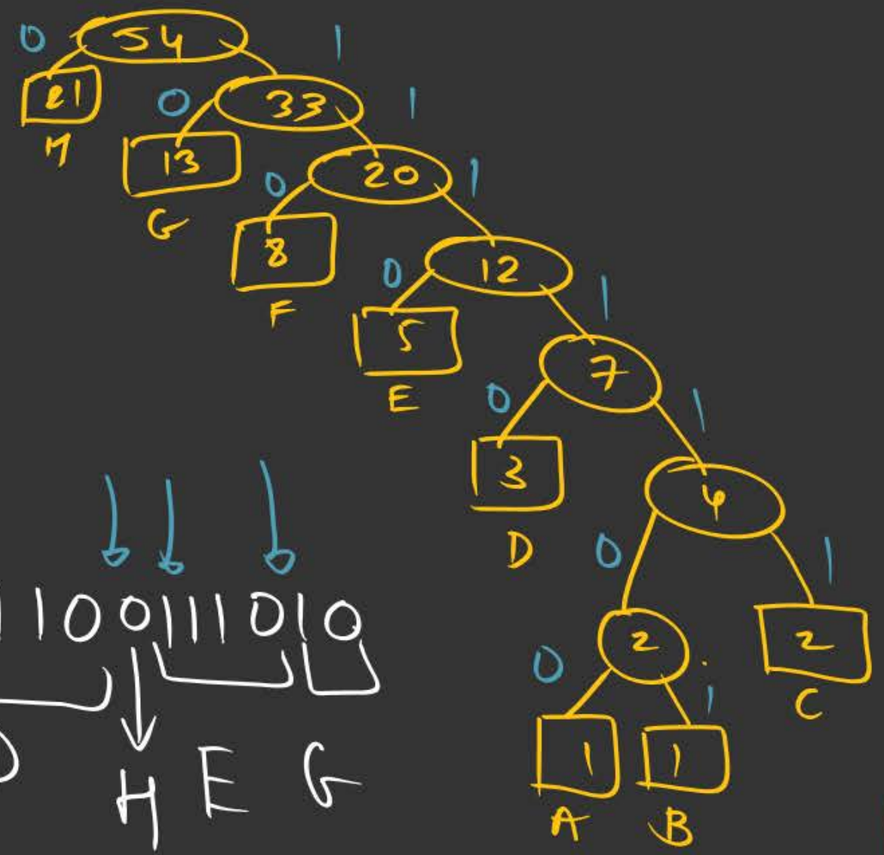
A FDHEG

B ECGDF

C DCHFG

D FEHDG

A = 1 — 1
B = 1 — 2
C = 1 — 3
D = 3 — 4
E = 5 — 5
F = 8 — 6
G = 13 — 7
H = 21 — 8



1 1 0 | 1 1 1 1 0 | 0 | 1 1 1 | 0 | 1 0

F D H E G

FDHEG

0 1 0

Root → leaf

A = 1 1 1 1 1 0 0

B = 1 1 1 1 1 0 1

C = 1 1 1 1 1 1

D = 1 1 1 1 0

E = 1 1 1 0

F = 1 1 0

G = 1 0 H = 0



Topic : Greedy Algorithms

PYQ

#Q. A message is made up entirely of characters from the set X = {P,Q,R,S,T}. The table probabilities for each of the characters is shown below:

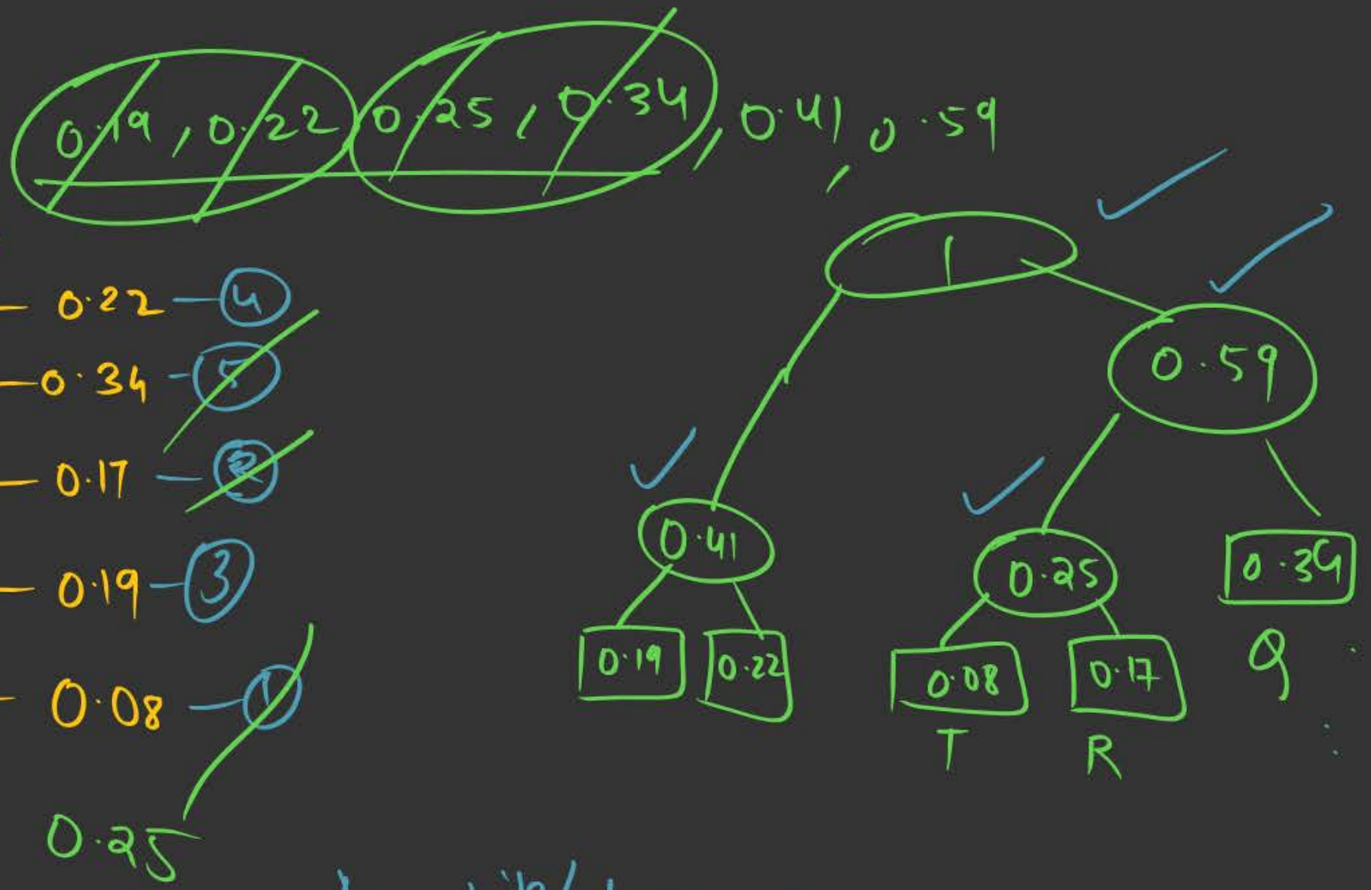
Characters	Probability
P	0.22
Q	0.34
R	0.17
S	0.19
T	0.08
Total	1.00

57.5%.

1) Avg no of bits/char

2) ans = 100 * avg.

If a message of 100 characters over X is encoded using Huffman coding, then the expected length of the encoded message in bits is_____.



for 100 char msg

Total bits $= 100 \times 2.25$

$= \boxed{225 \text{ bits}}$

Avg bit/char $= 1 + 0.59 + 0.41 + 0.25$

$= 2 + 0.25 = \boxed{2.25}$ bits/char

[MCQ]

PYQ

#Q. Consider the string abbccddeeee. Each letter in the string must be assigned a binary code satisfying the following properties:

1. For any two letters, the code assigned to one letter must not be a prefix of the code assigned to the other letter.
2. For any two letters of the same frequency, the letter which occurs earlier in the dictionary order is assigned a code whose length is at most the length of the code assigned to other letter.

Among the set of all binary code assignment which satisfy the above two properties, what is the minimum length of the encoded string?

A 25

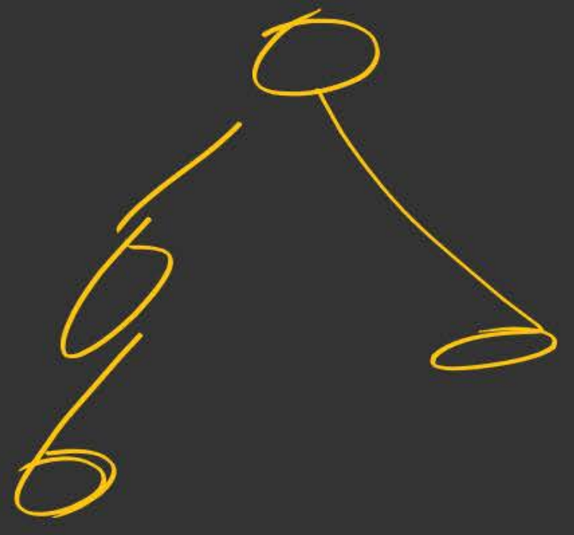
B 23

C 21

D 30

Sohil

txt = abbccddeeed



a = 1 — ①

b = 2 — ④

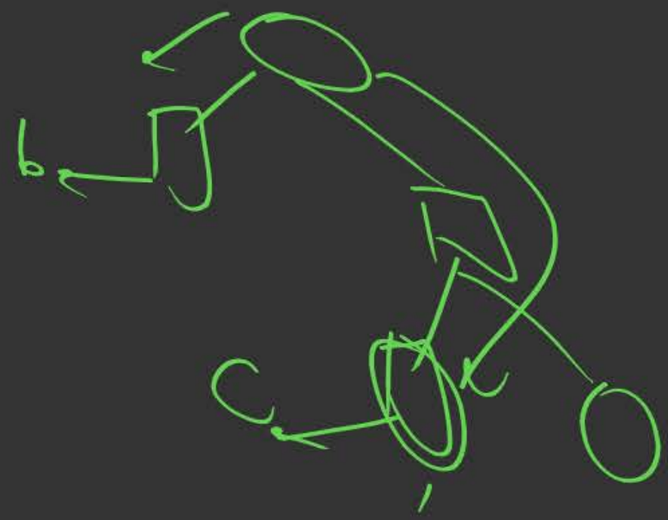
c = 2 — ③

d = 2 — ②

e = 3 — ⑤

$b \leq c \leq d$

$len(b) \leq len(c)$





Topic : Greedy Algorithm

Encoding:

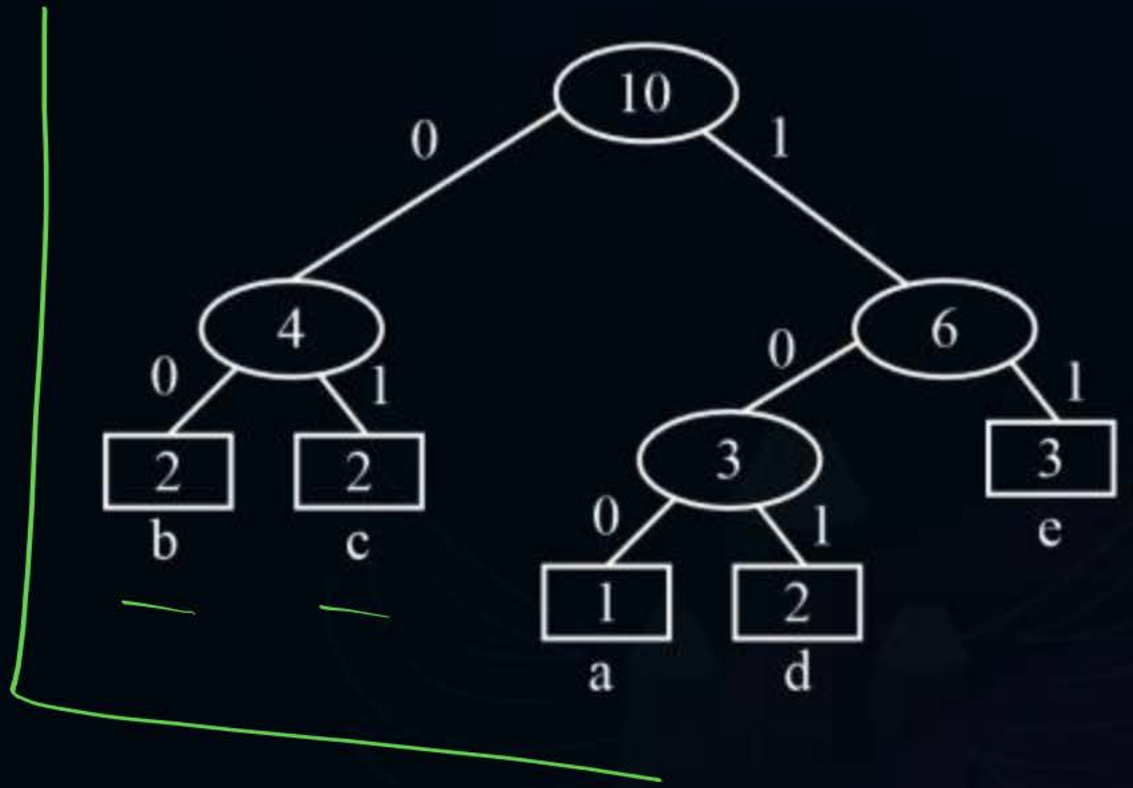
a → 100 (3)

b → 00 (2)

c → 01 (2)

d → 101 (3)

e → 11 (2)



Text "abbccddeee"

Total bits = 3 × 1 + 2 × 2 + 2 × 2 + 3 × 2 + 2 × 3

= 23

(V.V.V. Imp)

Min Cost Spanning Trees (MCST)



Topic : Greedy Algorithm

Minimum Cost Spanning Tree (MCST):

- Graph Based Problem
- Graph : G ≡ (V, E)

  where, V = Set of vertices

  E = Set of edges



Topic : Greedy Algorithm

Types:

(1) Undirected vs Directed:



Un_dir



dir

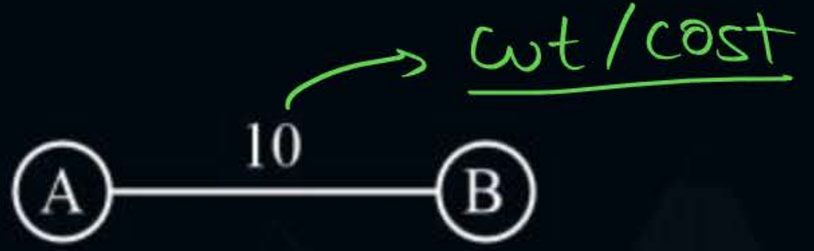


Topic : Greedy Algorithm

Types:

(2) Weighted vs Un-weighted:





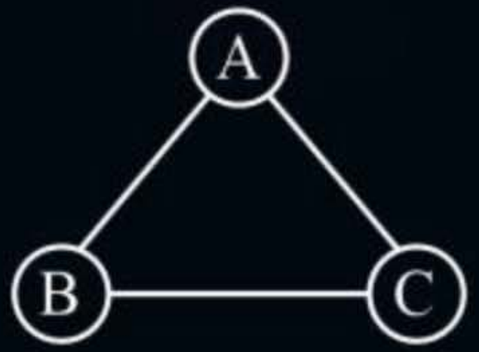


Topic : Greedy Algorithm

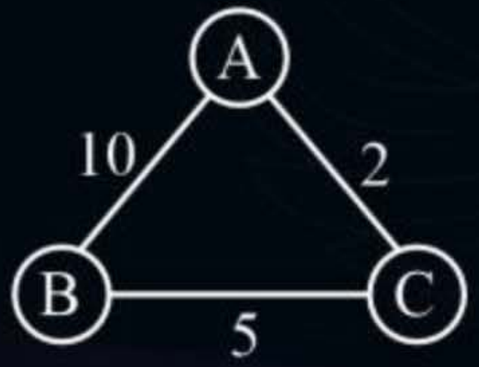
(1) Undirected Unweight Graph:

D W

UD UW



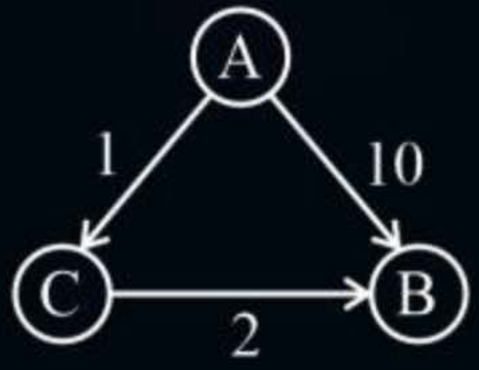
(2) Undirected Weight Graph:



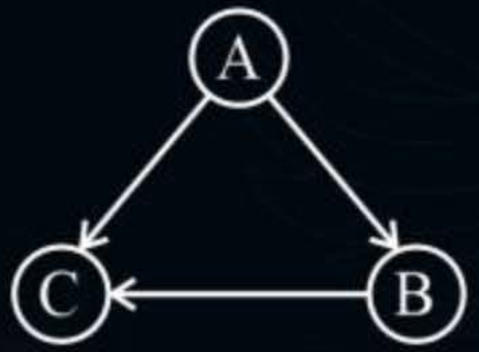


Topic : Greedy Algorithm

(3) Directed weight Graph:



(4) Directed Unweight Graph:





Topic : Greedy Algorithm

Representation of Graph:

(1) Adjacency matrix → (Array)

(2) Adjacency List → (List based)



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(1) Adjacency matrix:

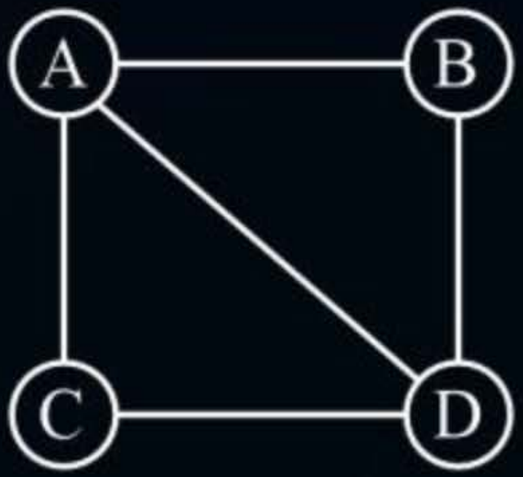
n = no. of vertices = 4

e = no. of edges = 5

Size = $O(n^2)$

Adj[i, j] = 1 $\quad (i, j) \in E$

Adj[i, j] = 0 $\quad (i, j) \notin E$



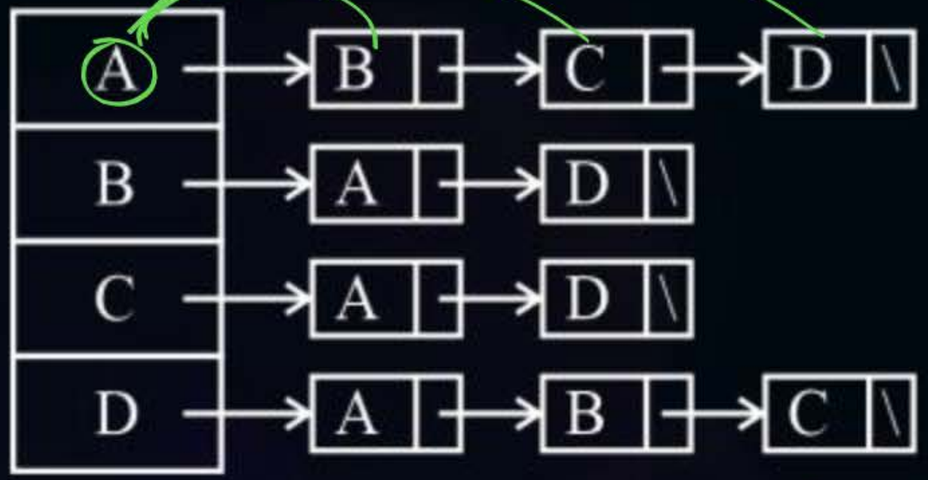
Adj	A	B	C	D
A	0	1	1	1
B	1	0	0	1
C	1	0	0	1
D	1	1	1	0

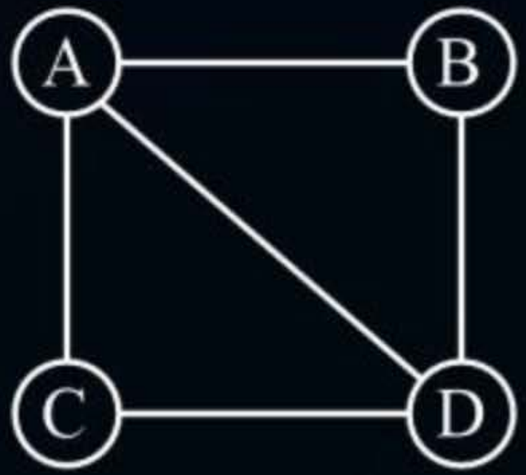


Topic : Greedy Algorithm

(2) Adjacency List:

Array of linked list





Size

(i) Undirected graph: (n + 2e)

(ii) Directed graph: (n + e)

→ O(n+e)



Topic : Greedy Algorithm

(1) Directed Complete Graph:

$e = n * (n - 1)$

$\rightarrow O(n^2)$

$V \rightarrow V$

$n(n-1)$





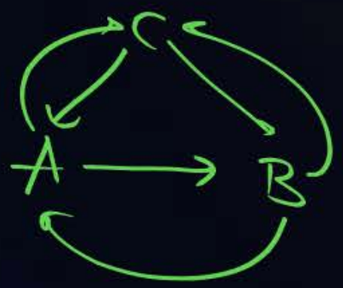
Topic : Greedy Algorithm

W.C (max) no. of edges

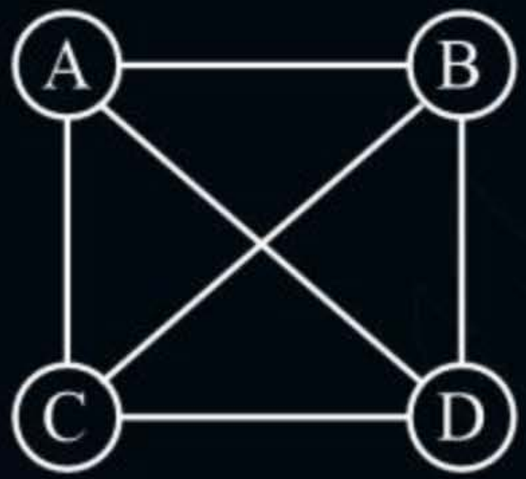
(2) Un-Directed Complete Graph:

$e = n(n-1)/2$

$\rightarrow O(n^2)$



A — B



[NAT]

#Q. Which representation to use when?

Adj Matrix → $O(n^2)$

Adj List → $O(n + e)$



Topic : Greedy Algorithm

Dense Graph:

Complete Graph:

- $e \approx O(n^2)$
- Matrix refer is preferred.

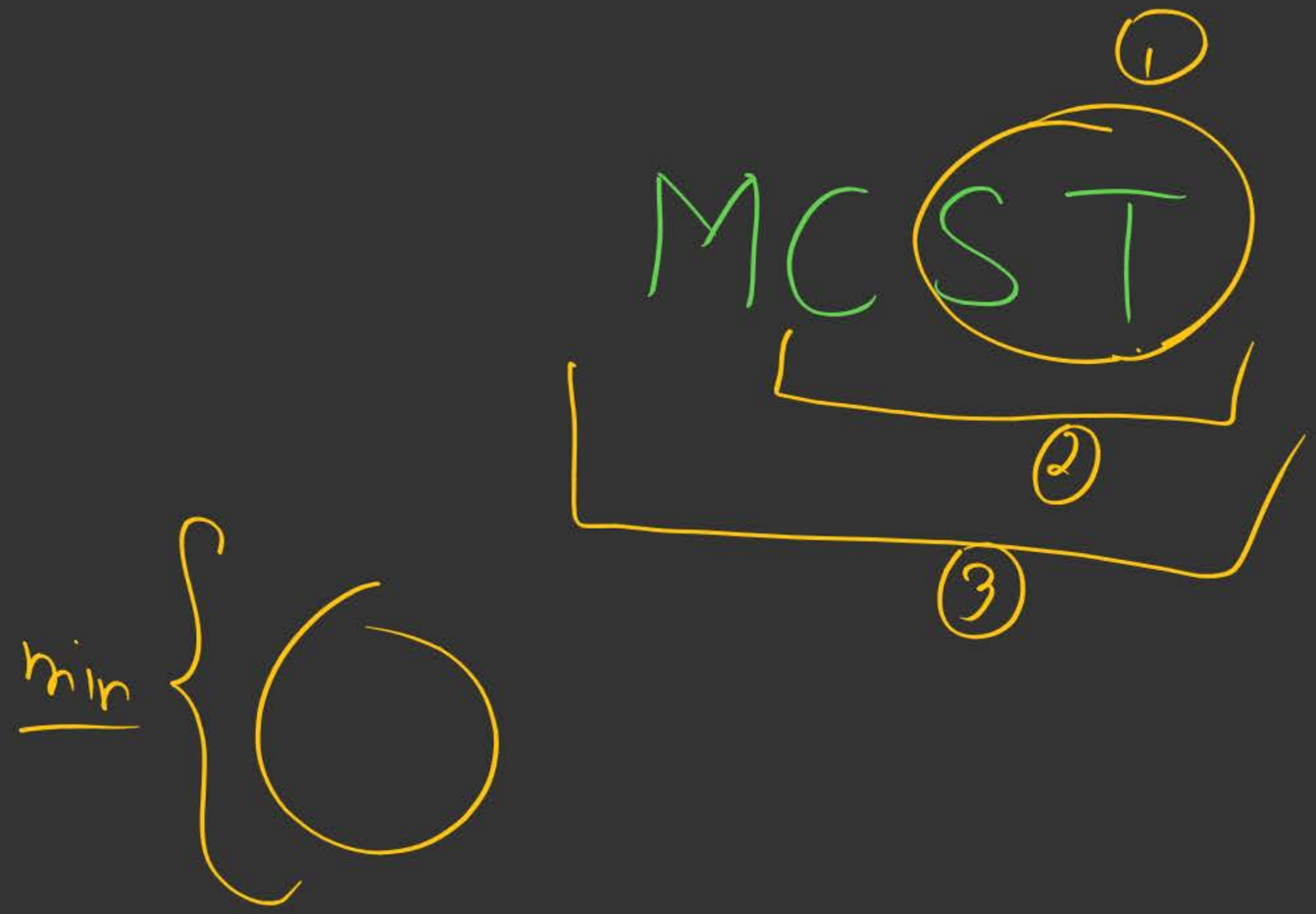
repr

Sparse Graph:

Non-Complete Graph:

usually $e \neq O(n^2)$

- Adj list is preferred.
- $O(n + e)$


MCST
1
2
3
min

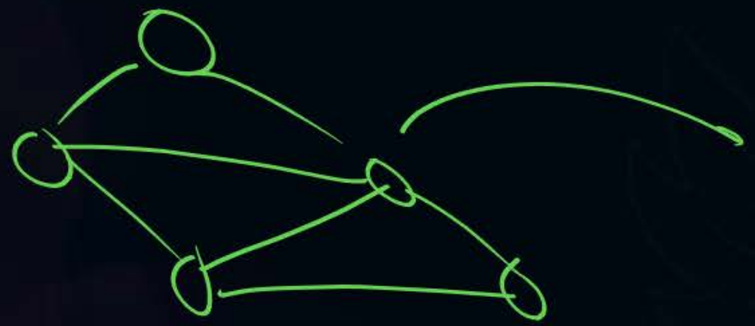


Topic : Greedy Algorithm

Spanning Tree:

- A subgraph T(V, E') of a given graph (V, E) where $E' \subset E$ is a spanning Tree iff T is a Tree.

Acyclic

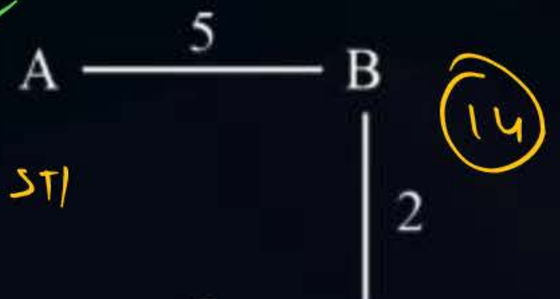


Topic : Greedy Algorithm

Example:

Q. Which of the following are valid Spanning Tree of G?

A

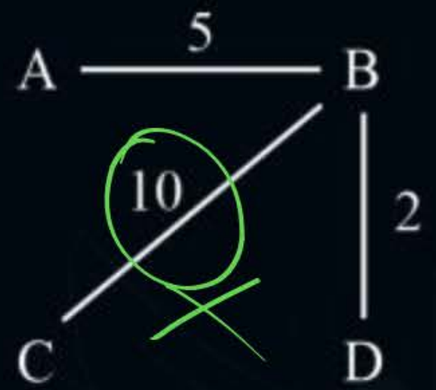


ST1

14

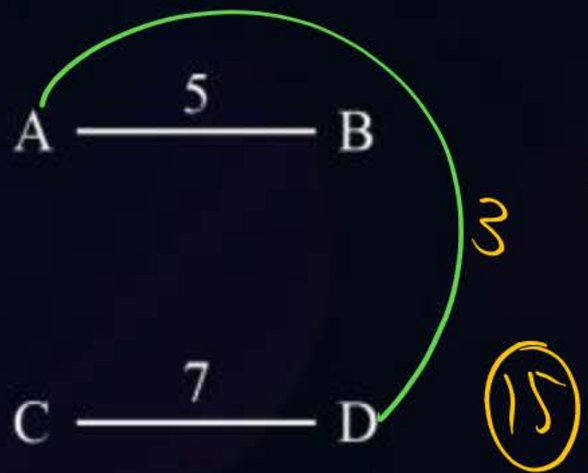
B

ST2



C

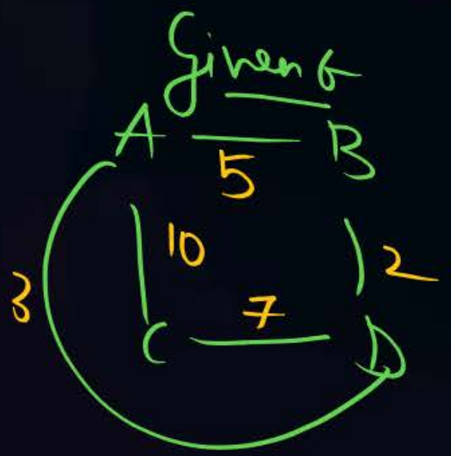
ST3



3

15

Given G





Topic : Greedy Algorithm

Important Observations:

1) Multiple Spanning Trees are possible

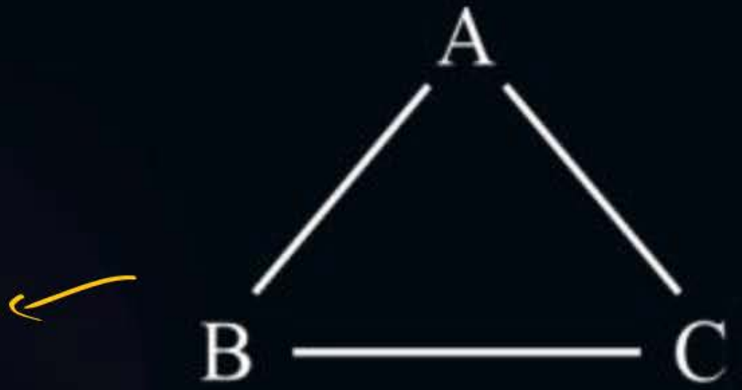
ST – 1 → 5 + 2 + 7 = 14

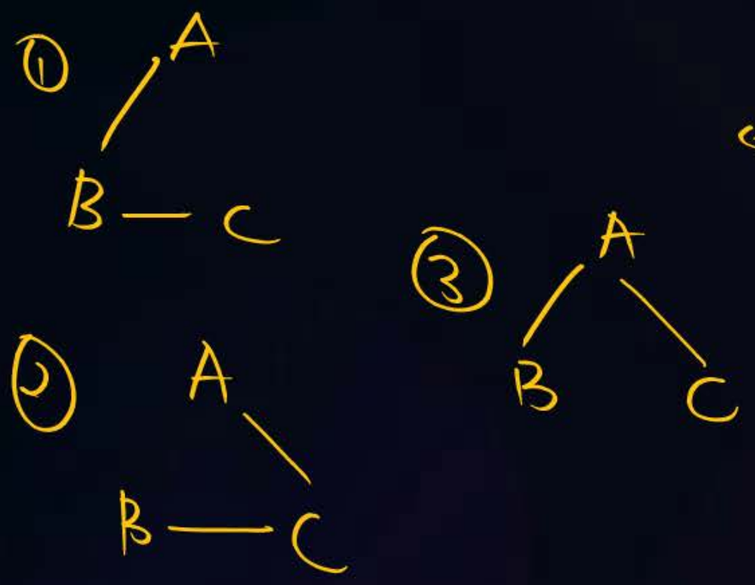
ST – 2 → 5 + 3 + 7 = ~~14~~ 15

2) A Spanning Tree with ~~n-matrices~~ n-vertices will always have (n – 1) edges.

[NAT]

#Q. For a given graph with n-vertices and e-edges the maximum number of Spanning Tree =?





$\Rightarrow$ $n^{(n-2)}$ Spanning Trees

Brute force

$= \frac{n^n}{n^2} \approx O(n^n)$

[NAT]

#Q. Graph: n → vertices,

e → edges

The no. of edges that must be removal from this graph to get its Spanning Tree?

↳ Spanning Tree ⇒ $e' = (n-1)$



Topic : Greedy Algorithm

- Spanning Tree of 'n' vertices → (n – 1) edges required.

Total edges = (required + removal)

↓ ↓ ↓

e = (n–1) + x

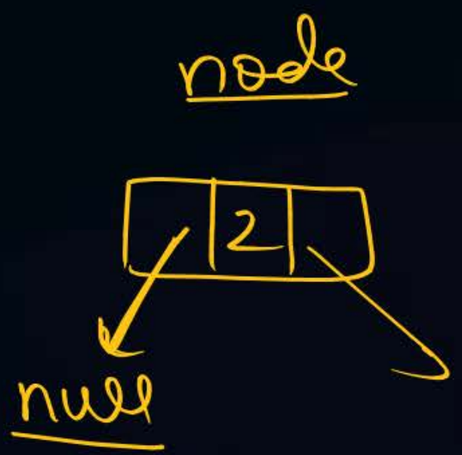
x = e – (n – 1)

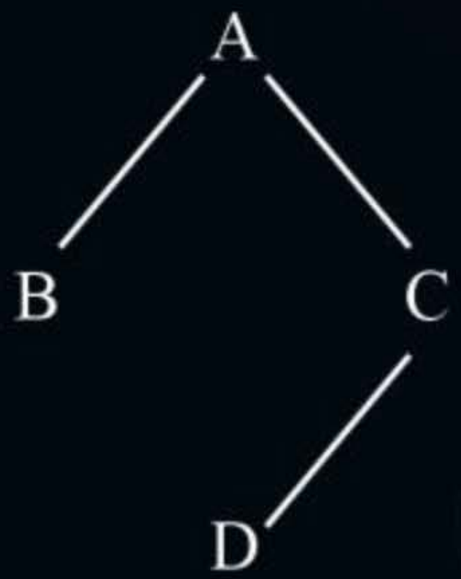
x = e – n + 1

e' = (n-1)

[NAT]

#Q. A Tree with 'n' nodes/vertices will have _____ no. of null links ?







Topic : Greedy Algorithm

Logic: Total links in a tree with n vertices? = 2 * n

edges (used) = (n – 1) → non-null links

unused/null-links = Total – edges (used)

= 2n – (n – 1)

= 2n – n + 1

= n + 1





2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Huffman
PYQs
MCST



THANK - YOU